

ROAD-SENSE

THE ROAD-HEALTH INTELLIGENCE LAYER



Class 12 Technical Presentation | Hardware Prototype + Edge AI + Satellite Geolocation Architecture

THE ROAD DEGRADATION PROBLEM

Systemic Infrastructure Challenge

⚠ Current Bottlenecks

- ▶ **Manual Inspection Lag:** Inspection relies on visual surveys or reactive citizen complaints.
- ▶ **Lack of Spatial Precision:** Standard GPS has high drift, obscuring exact lane-level defect points.
- ▶ **Unstructured Data:** Photos uploaded by citizens lack standardized depth, severity, or structural metrics.
- ▶ **Reactive Maintenance:** Potholes are fixed only after severe escalation, multiplying repair costs.

🔄 The Traditional Process

TRAVEL → NOTICE DEFEAT → MANUAL REPORT → DEPT VERIFICATION →
PRIORITY ALLOCATION (Weeks / Months)

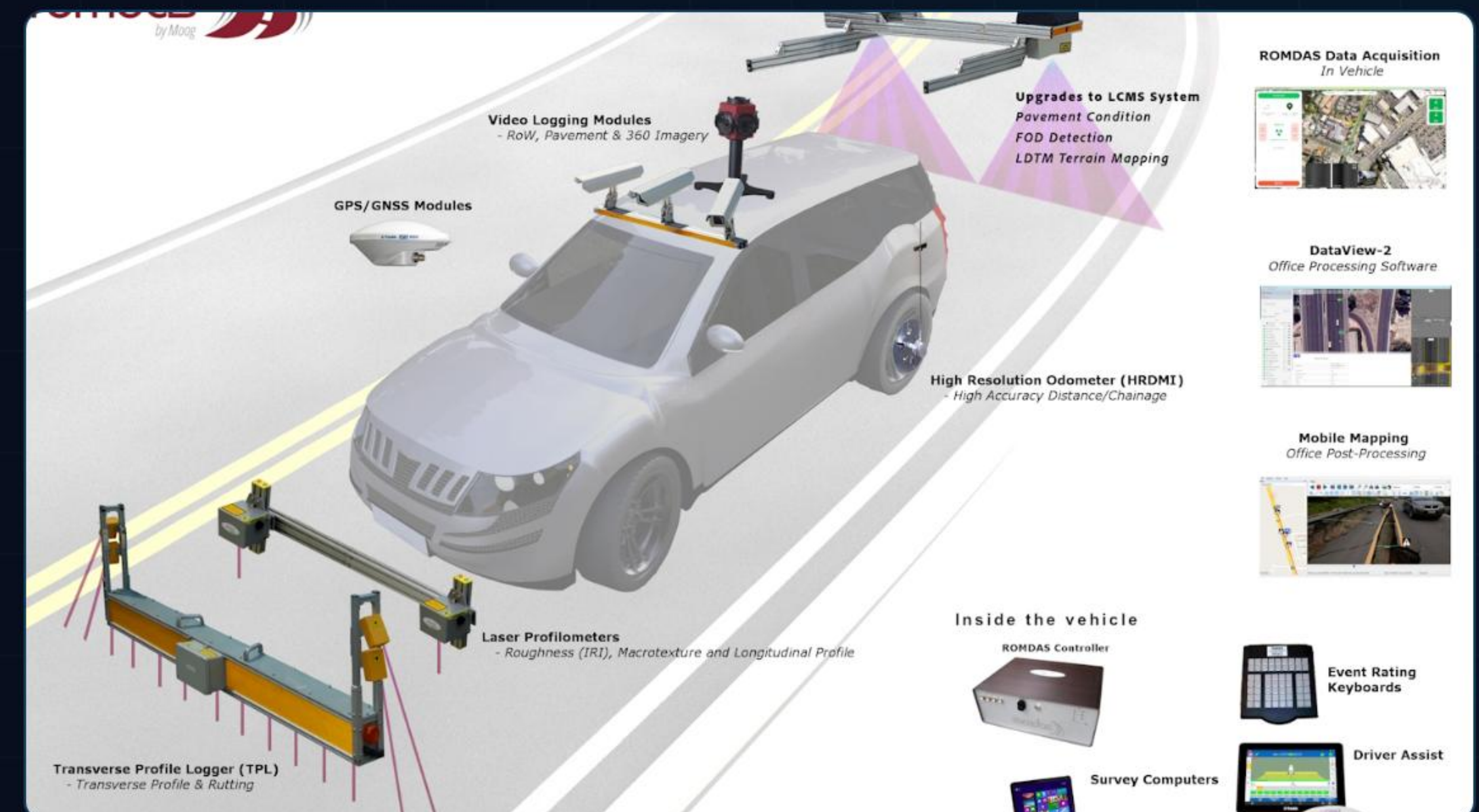


Fig 2.1: Conventional road monitoring lacks continuous, high-frequency digital sensing.

THE INNOVATION: PASSIVE SENSING

Paradigm Shift

✗ Traditional Citizen/Manual Reporting

 Human Notice Required

 Manual App Opening & Form Filling

 Subjective Evaluation & Delay

✓ ROAD-SENSE Passive Intelligence

 Automatic Journey Sensing during routine transit

 Edge Sensor Fusion (IMU + Vision + NavIC)

 Structured Road-Segment Health Score Generated

"ROAD-SENSE transforms every routine vehicle trip into an automated, structured road quality audit."

SYSTEM CONCEPTUAL MIND MAP

8 Core Pillars

1. SENSORS

- Wide-Angle Road Camera
- 6-DOF IMU (Accelerometer/Gyro)
- Ambient Surface Temp Sensor

2. POSITION

- NavIC L1/L5/S Band Engine
- Sub-meter GNSS Interpolation
- Nanosecond Atomic Timing

3. ELECTRONICS

- ESP32 Sensor Hub Board
- Raspberry Pi-class Edge SBC
- Regulated Power Management

4. INTELLIGENCE

- Edge Computer Vision
- Dynamic Motion Z-Spike Detector
- Multi-modal Sensor Fusion

5. DATA LAYER

- Road Segment ID Hashing
- Pothole/Crack Classification
- Observation Repeat Counter

6. OUTPUT

- Segment Health Index (0-100)
- GIS Priority Heatmap
- Municipal Action Dashboard

7. ECOSYSTEM

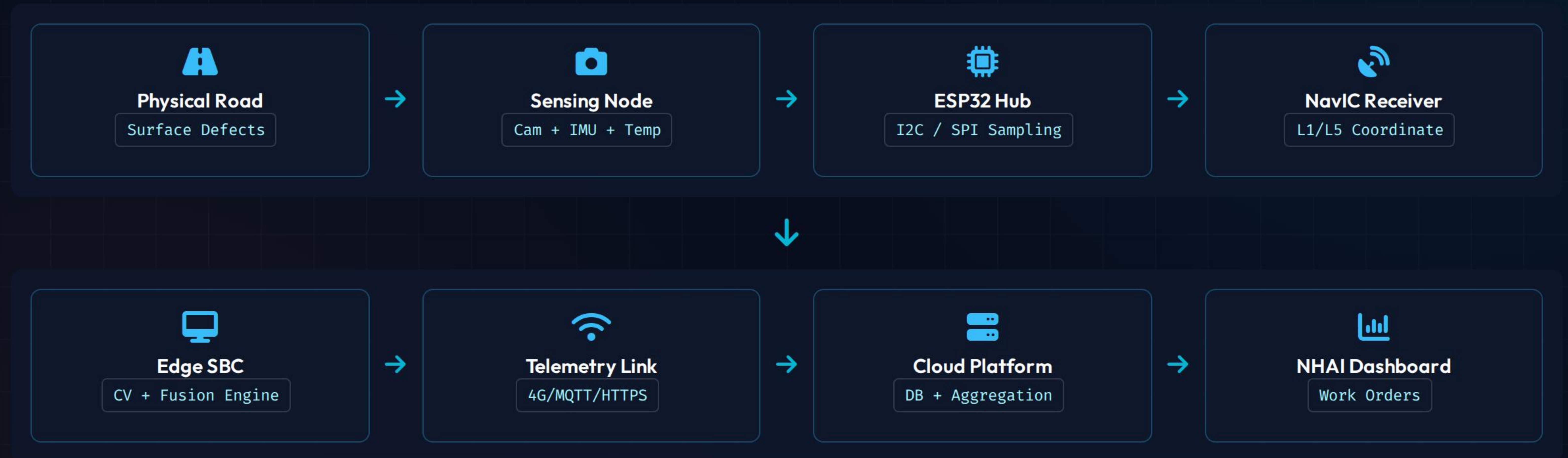
- ISRO NavIC Constellation
- NISAR Satellite Cross-ref
- PWD / NHAI Infrastructure

8. SCALING

- Bench Prototype (L1)
- Fleet Integration (L3)
- National Road Grid (L5)

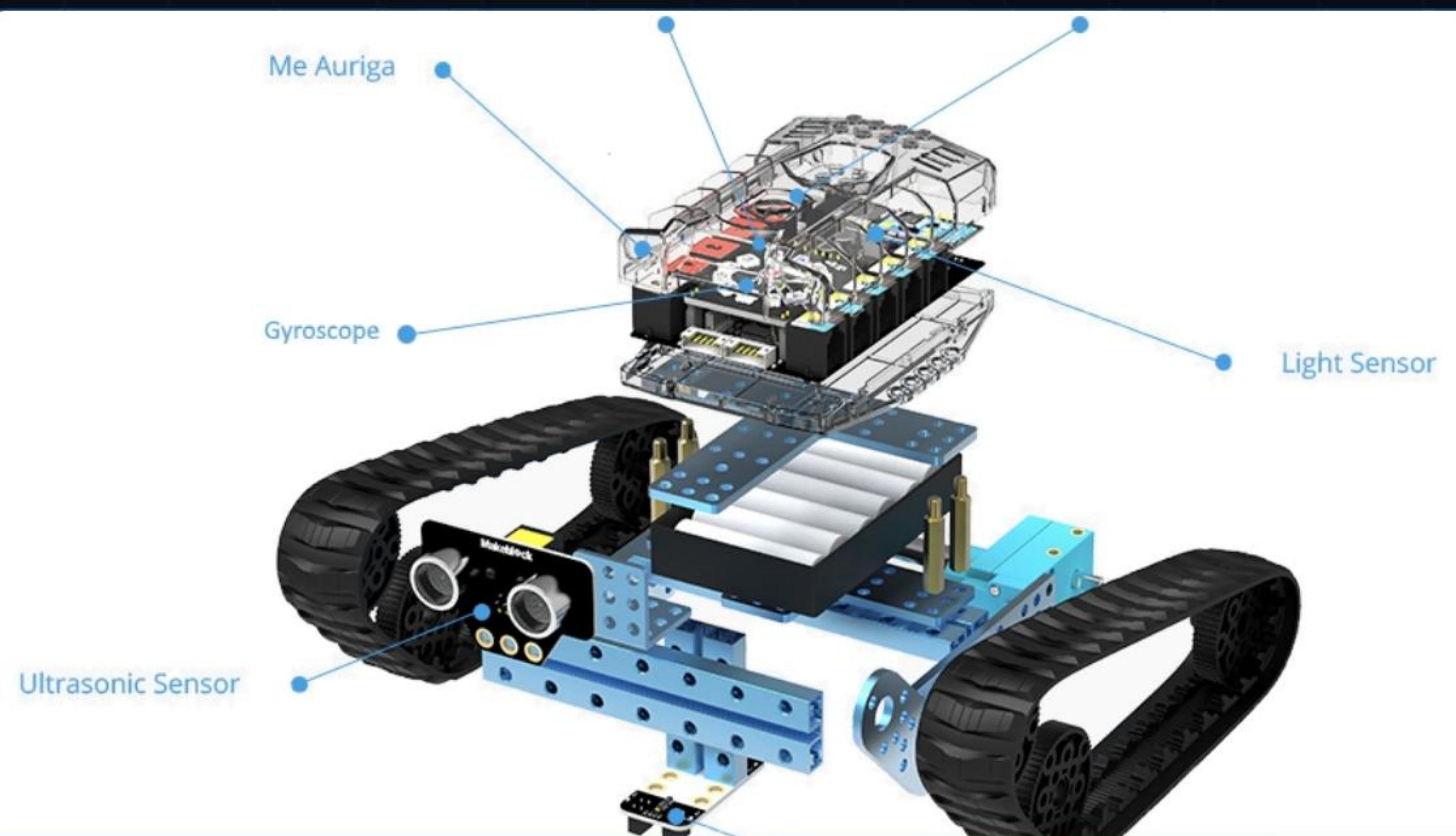
END-TO-END SYSTEM ARCHITECTURE

Hardware to Cloud Flow



FIRST PHYSICAL PROTOTYPE PLATFORM

Safe Bench Testbed



Controlled Bench Platform

Mounted inside a rigid acrylic/3D-printed enclosure attached to a controlled test vehicle or tabletop track setup.

Subsystem Layout

- ▶ **Camera:** Front-facing wide field of view.
- ▶ **ESP32 Sensor Hub:** High-speed IMU sampling.
- ▶ **NavIC Antenna:** Top-mounted ceramic patch antenna.
- ▶ **Edge SBC:** Local processing & USB storage.
- ▶ **Power Supply:** Regulated 5V 3A step-down module.

PROTOTYPE SCALE PROGRESSION

5-Phase Deployment

LEVEL 1

Bench Rig

Tabletop test track & artificial vibration plate.

LEVEL 2

Vehicle Rig

Supervised single vehicle test drives on campus.

LEVEL 3

Fleet Pilot

Deployment on 10 school buses / municipal vans.

LEVEL 4

City Grid

500+ public transport & logistics vehicles.

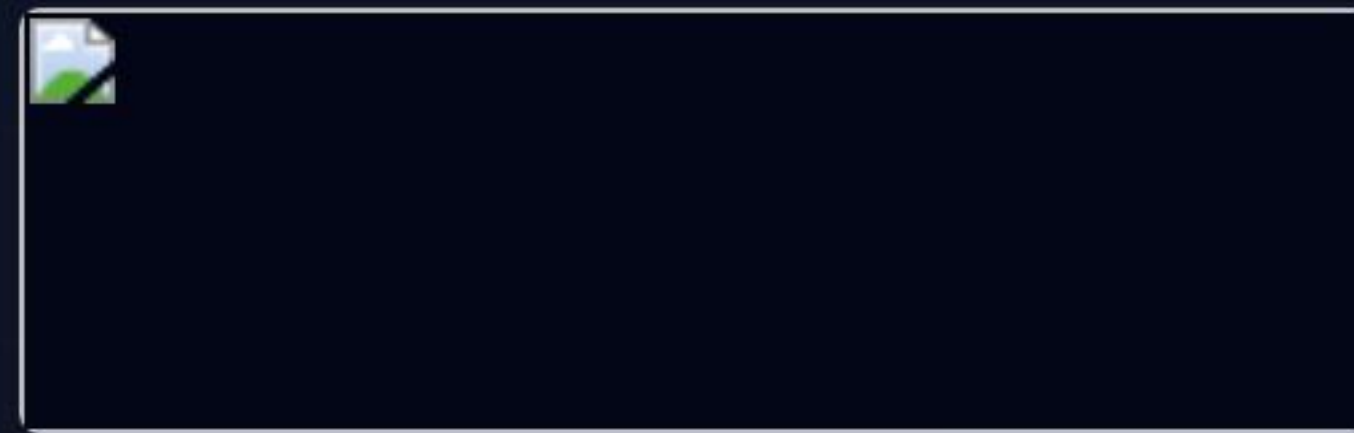
LEVEL 5

State Network

Statewide continuous road quality telemetry.



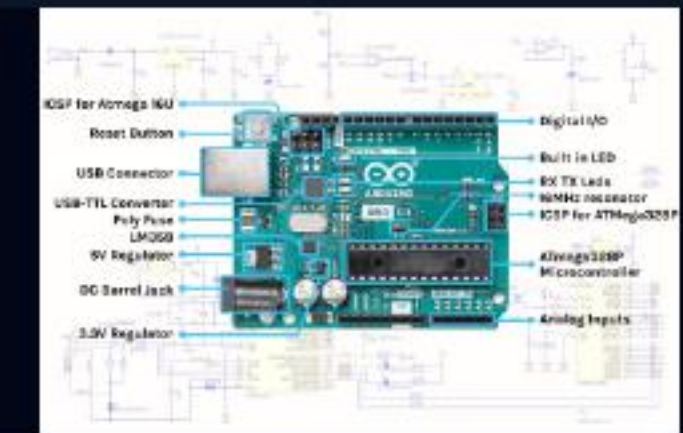
ESP32 Node
Sensor Sampling Hub



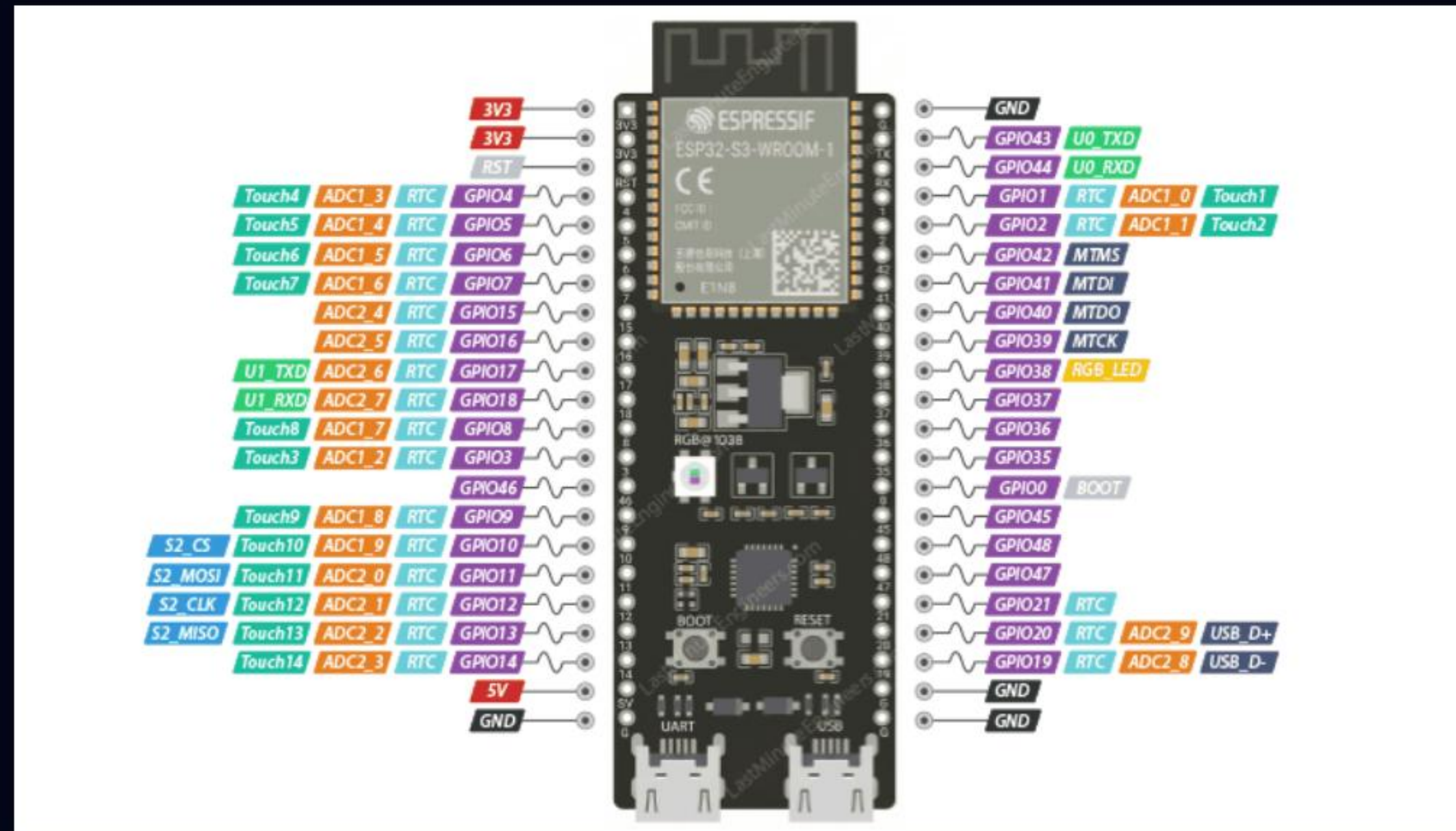
Raspberry Pi 4
Edge Vision Engine



NavIC GNSS
ISRO L1/L5 Receiver



6-DOF IMU
Z-Axis Acceleration



🔧 Dual-Core 240MHz MCU

- ▶ **Primary Function:** Dedicated real-time sensor sampling (200Hz IMU read rate).
- ▶ **Interfaces:** I2C for IMU/Temp sensor, SPI for SD logging, UART to GNSS.
- ▶ **Role Distinction:** ESP32 handles low-latency real-time interrupts, leaving heavy computer vision tasks to the SBC.

Arduino Uno (Optional Lab Tool)

- ▶ Used in early bench testing for basic sensor verification.
- ▶ Limited 8-bit ATmega328P CPU (16MHz, 2KB RAM).
- ▶ **Status:** Not used in main production system due to lack of memory and Wi-Fi/edge bandwidth.

ESP32 + SBC Combination

- ▶ 32-bit dual core with integrated Wi-Fi / Bluetooth stack.
- ▶ Direct high-speed buffer sharing with Raspberry Pi Edge Engine.
- ▶ **Status:** Core architecture standard for ROAD-SENSE.

 Raspberry Pi

Edge Computing Principles

- ▶ Processes raw camera video streams in real-time on-board.
- ▶ Executes quantized YOLO/MobileNet defect classification models.
- ▶ **Bandwidth Saver:** Transmits only telemetry meta-data (coordinates, defect level) to cloud rather than raw video.

VISUAL DEFECT DETECTION CAMERA

Optical Intelligence

Optical Specifications

- ▶ 1080p @ 60 FPS wide-angle camera with global shutter option.
- ▶ Calibrated optical perspective transform to measure surface area.
- ▶ Identifies: Potholes, alligator cracking, rutting, and longitudinal cracks.



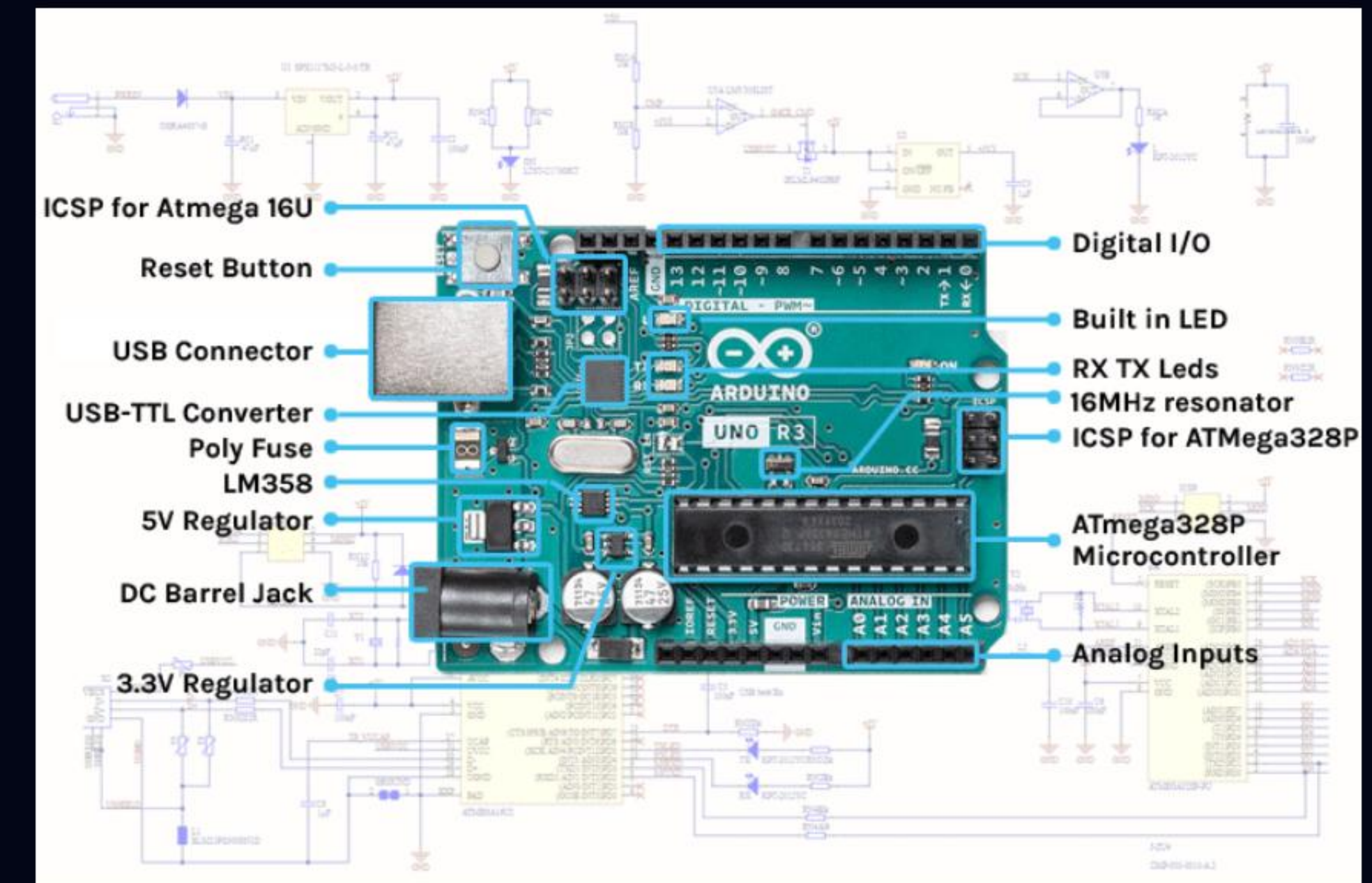
6-DOF IMU INERTIAL MOTION SENSING

Z-Axis Acceleration Spikes

Accelerometer Mathematical Signal

$$A_{\text{mag}} = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

When a vehicle hits a pothole, a sharp impact impulse appears in the vertical Z-axis acceleration profile.



Bitumen Temperature Factor

- ▶ Asphalt stiffness and durability vary drastically with surface temperature.
- ▶ High thermal expansion during summer causes rutting; thermal contraction leads to transverse cracking.
- ▶ Temperature metadata contextualizes why road degradation is accelerating in specific regions.

Infrared Non-Contact Sensor

I2C IR Thermometer

Measures asphalt surface temperature independently of vehicle chassis warmth.



✦ Strategic Advantages of NavIC

- ▶ **Dual Band (L5 & S-Band):** Multi-frequency operation offers superior positioning in urban canyons and forested roads.
- ▶ **Next-Gen NVS Series (L1 Band):** Compatible with standard low-cost GNSS receivers.
- ▶ **Indigenous Accuracy:** Sub-5 meter positioning over India and 1,500 km surrounding landmass.

Vehicle 12V Input

Takes raw vehicle power (9V-36V DC) via OBD-II or cigarette lighter port.

Regulated Step-Down

Buck converter outputs ultra-stable 5V @ 4A with over-voltage and surge protection.

Backup LiFePO4

Uninterruptible power buffer prevents SD card corruption during engine ignition dips.

Offline Resiliency

- ▶ Vehicles frequently travel through cellular blind spots in rural India.
- ▶ High-endurance micro-SD / NVMe SSD buffers logged telemetry locally in SQLite database.
- ▶ Automatic sync occurs whenever 4G/5G or depot Wi-Fi reconnects.

Fail-Safe Data Logging

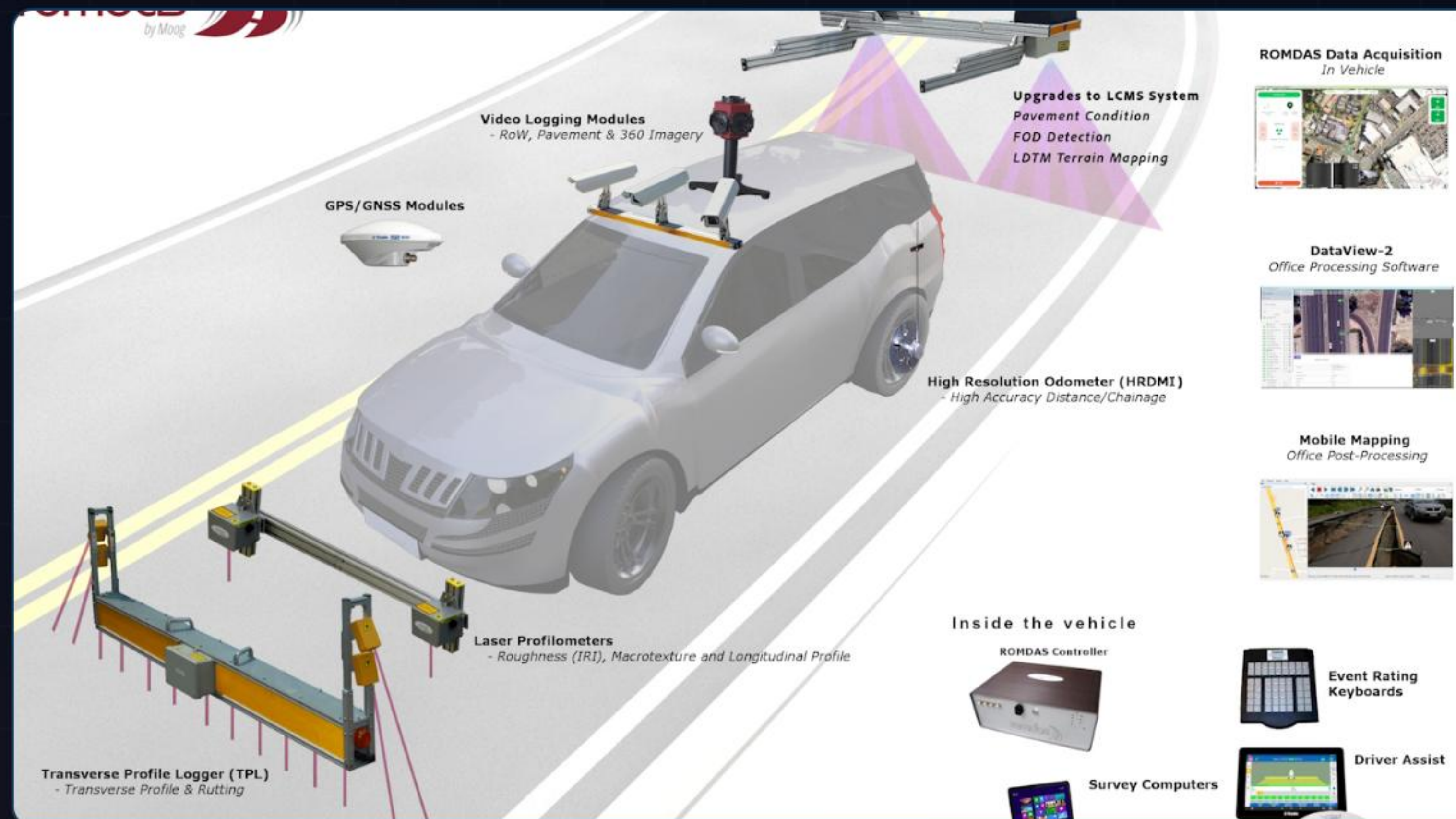
Circular FIFO Queue

AES-128 Encryption

HARDWARE WIRING SCHEMATIC

Inter-Module Bus Topology





👁 Vision Processing Steps

- ▶ 1. Frame extraction & perspective correction.
- ▶ 2. Bounding box detection via quantized neural network.
- ▶ 3. Class identification: Pothole vs Crack vs Patch.
- ▶ 4. Area measurement approximation in sq cm.

Spike Peak Detection

A true pothole produces a combined sharp negative Z-axis displacement followed by a high-g acceleration rebound peak.

$$Z(t) > T_{\text{threshold}}$$

Speed Normalization

Vibration intensity scales with vehicle velocity. High-speed minor bumps must not be false-classified as potholes.

Fused Confidence Scoring Matrix

Camera ONLY

Possible shadow / puddle false positive.

Confidence: 40%

IMU ONLY

Possible speed bump or expansion joint.

Confidence: 50%

FUSED (Cam + IMU)

Simultaneous optical shadow & Z-impact spike.

Confidence: 95%

Road Health Severity Equation

$$S = w_1 \cdot V_{\text{cam}} + w_2 \cdot I_{\text{imu}} + w_3 \cdot f(T)$$

Where V represents visual severity, I represents inertial spike magnitude, and f(T) adjusts for temperature-induced bitumen vulnerability.

```
{  
  "segment_id": "NH19_KM_412",  
  "navic_lat": 26.4499,  
  "navic_lon": 80.3319,  
  "timestamp_utc": "2026-09-03T10:30:00Z",  
  "imu_z_peak": 2.41,  
  "vision_defect_class": "pothole",  
  "confidence": 0.94,  
  "surface_temp_c": 42.5  
}
```

Spatial Hashing

Roads are divided into 10-meter discrete segments using spatial indexing, allowing multiple vehicle observations to aggregate seamlessly over time.



Real-Time Heatmap Features

- ▶ **Red Segments:** Critical potholes requiring immediate intervention within 48h.
- ▶ **Yellow Segments:** Developing cracks; scheduled for preventative sealing.
- ▶ **Green Segments:** Healthy road surface.

NavIC Positioning Layer

Provides micro-level ground truth vehicle coordinates during edge sensing.

NISAR Radar Remote Sensing

Complements ground vehicle sensing with satellite Synthetic Aperture Radar (SAR) to monitor ground subsidence and major highway erosion.

NATIONAL ROADWAY SCALING STRATEGY

From IIT Kanpur to All-India

PHASE 1

Campus Validation

IIT Kanpur campus shuttle fleet deployment.

PHASE 2

Kanpur City

100 Municipal buses & garbage trucks.

PHASE 3

Expressways

Yamuna Expressway & NH-19 corridor fleets.

PHASE 4

National Grid

Integration with NHAI PM Gati Shakti Portal.

B2G (Government / Municipal)

Subscription model for NHAI, PWD, and Smart City corporations for automated road health auditing and contractor quality verification.

B2B (Logistics & Insurance)

Provides optimal route planning for commercial logistics to minimize vehicle wear, damage, and tire blowouts.

Safety First Protocol

Testing is restricted to controlled rigs and normal driving speeds. Prototype testing NEVER involves creating or driving recklessly into hazards.

Privacy by Design

Edge computer automatically blurs faces and license plates prior to telemetry transmission, complying with DPDP Act 2023.

TECHNICAL SUMMARY & KEY ADVANTAGES

Why ROAD-SENSE Wins

Edge Intelligence

Zero cloud bandwidth lag; real-time sensor fusion directly at vehicle node.

NavIC Precision

Sub-meter accurate position metadata natively tailored for Indian geography.

Frugal Engineering

Low-cost hardware BOM suitable for mass deployment across public transit.

 FUTURE OF INDIAN INFRASTRUCTURE

THANK YOU

ROAD-SENSE: The Road-Health Intelligence Layer

Open for Questions & Technical Demonstration | IIT Kanpur YEC 2026

IMAGE SOURCES



<https://romdas.com/images/romdas-system-diagram.jpg>

Source: romdas.com



https://core-electronics.com.au/media/wysiwyg/tutorials/patrick/makeblock/ranger/mbot-ranger_sensors.png

Source: core-electronics.com.au



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Source: circuitdigest.com

IMAGE SOURCES



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Source: www.pmfias.com



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Source: www.supermap.com